Lab 3: Architectural Pattern Selection – Written Justification

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Assigned scope: Privacy, scorecard handling and soft-deletion (FR-005, NFR-001).

We chose Layered Architecture for the assigned subsystem of the Mock Interview and Alumni Mentorship Platform.

Style	Pros	Cons	Fit for this scope
Layered	Clear separation, central security checks	Overhead per layer; scales as one block	Best: strong privacy control, simple
Client-Server	Central control, easy consistency	Single point of failure	Acceptable but no layer-wise checks
Microservices	Independent scaling, fault isolation	Data split across services; complex	Weak: soft-deletion across services is hard

Architectural Choice

Layered Architecture.

Two Reasons

1. Privacy rules must be applied the same way everywhere. A Business layer between the Portal and the Repository gives one place (Access Control & Privacy Manager) where every request for profile or scorecard data is checked, so no feature can reach the data without passing it.

2. Soft-deletion needs strong data consistency. With one Repository in the Data layer, a deletion request marks profile and scorecard records as deleted in one consistent step within the 24–hour limit, and the layered structure is simple for a small team to build and test.

Security Advantage

Access is allowed only through the layers: the UI cannot query the database directly, and each Business-layer call goes through the IAccessCheck interface (owner-only checks and audit log). Unauthorized users, such as a second account trying to read another user's scorecard, are rejected before data is touched.

Performance Benefit

The workload (recording scorecards, handling deletion requests) is light and not peak-driven, so the small overhead of passing through layers is acceptable. Keeping the data in one repository also avoids network hops between services, which keeps scorecard reads and writes fast.